


*Faculty Of Engineering and Technology*

*Electrical and Computer Engineering Department*

*ENCS3330*

*Digital Integrated Circuits*

*Assignment #2*

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**Prepared by : Rahaf Salahat**

**ID: 1231584**

**Instructor: Dr. Khader Mohammad**

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# 1. D_Latch

The D Latch is a level-sensitive storage element used to store one bit of data. It has two main inputs: the data input (D) and the control signal (CLK or Enable). When the clock signal is active, the output follows the input D. When the clock signal becomes inactive, the latch stores the last input value and maintains it at the output. D Latches are commonly used as the basic building blocks for sequential circuits and flip-flops.

## 1.1 D_Latch Schematic

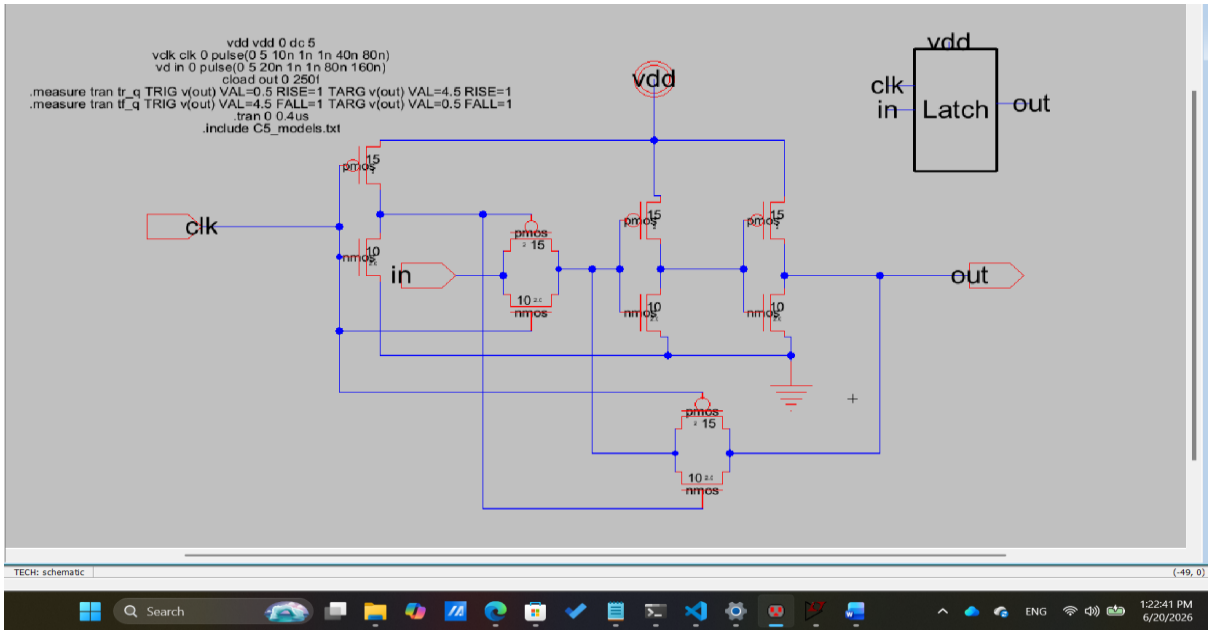



*Figure 1:D_Latch schematic*

## 1.2 D_Latch Layout

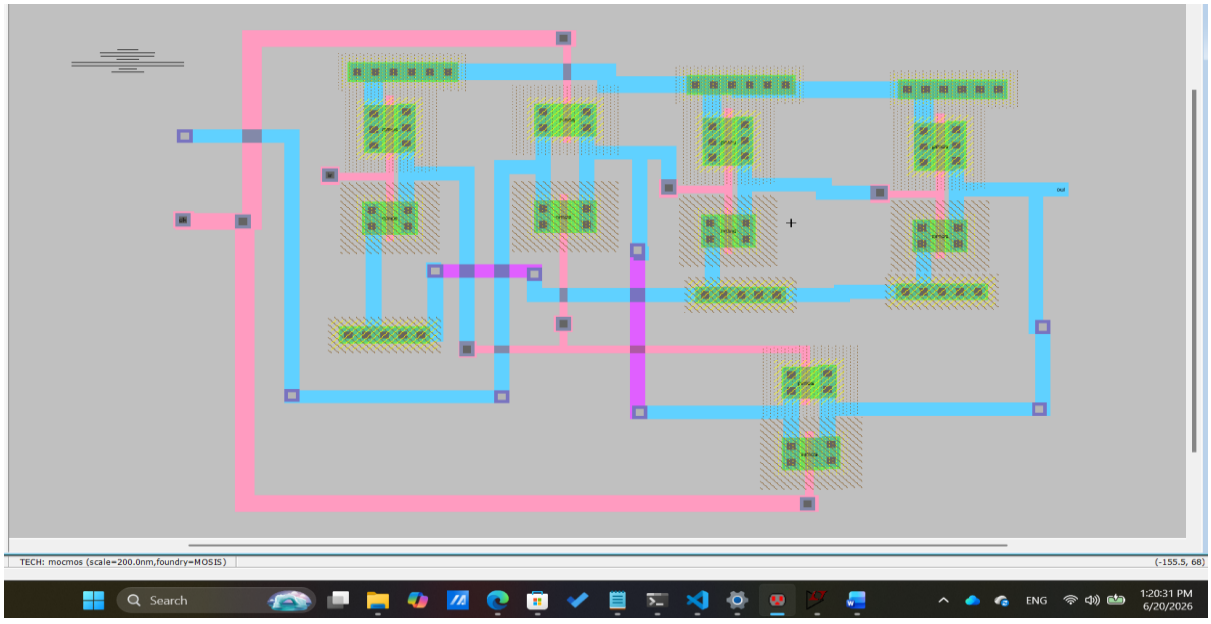


*Figure 2:D_Latch Layout*

The D Latch layout was designed in Electric VLSI using CMOS design rules, with PMOS transistors connected to VDD and NMOS transistors connected to GND. Metal-1 was used for routing connections between transistors, and DRC checks were performed to eliminate spacing and overlap errors. The final layout successfully represents the schematic and provides a valid CMOS implementation of the D Latch.

## 1.4 D_Latch Schematic

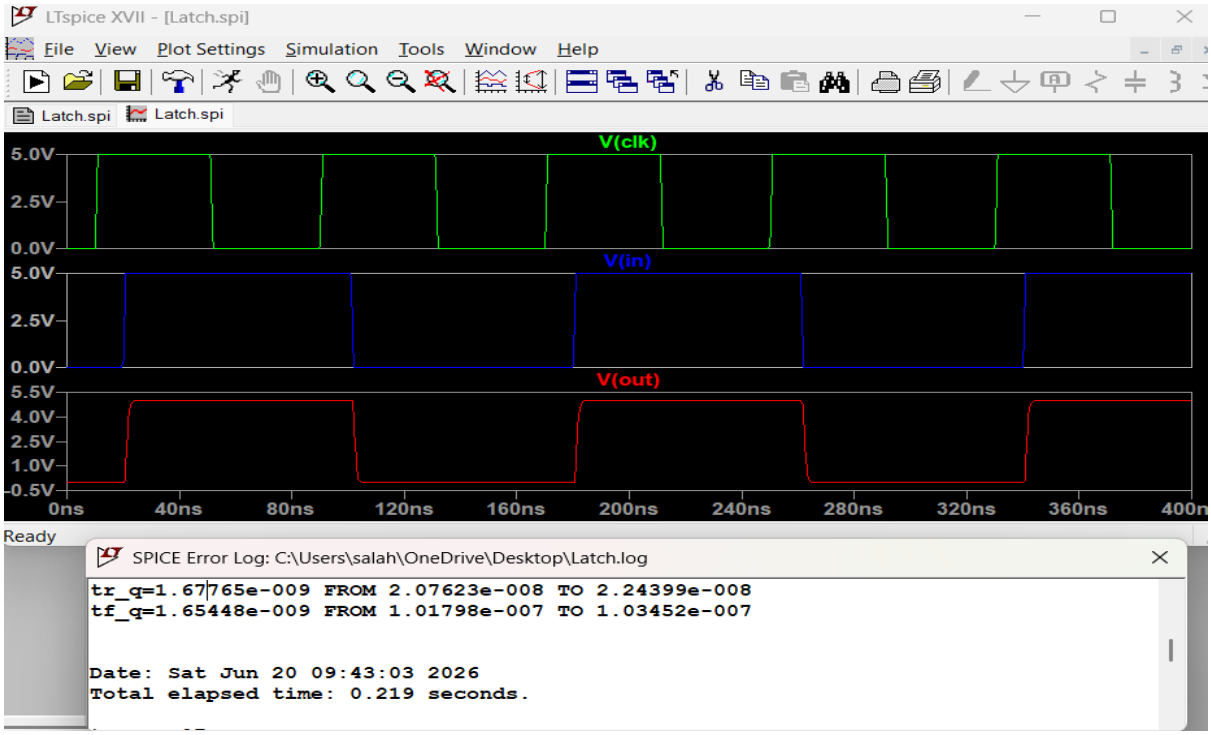



*Figure 3:D_Latch Simulation*

## 1.4 Discussion

The D Latch was simulated using LTspice with a 5V supply voltage. The input signal (D) and clock signal (CLK) were applied as pulse waveforms to verify the functionality of the circuit. The simulation results show that the output follows the input when the clock is active and retains the stored value when the clock becomes inactive, which confirms the correct operation of the D Latch.

The measured rise time and fall time of the output were approximately:

- Rise Time (tr) = 1.68 ns
- Fall Time (tf) = 1.65 ns

These values indicate fast switching performance and demonstrate that the latch responds correctly to changes in the input signal during the active clock period. The waveform verifies that the circuit successfully stores and maintains data as expected.

## 2. Falling Edge Flip Flop

A Falling Edge D Flip-Flop is a sequential storage element that captures the input data (D) only when the clock signal transitions from high to low (falling edge). It is commonly implemented using two D Latches connected in a master-slave configuration with complementary clock signals. This design ensures that the output changes only at the falling edge of the clock, providing reliable and synchronized data storage.

### 2.1 Falling Edge Flip Flop Schematic

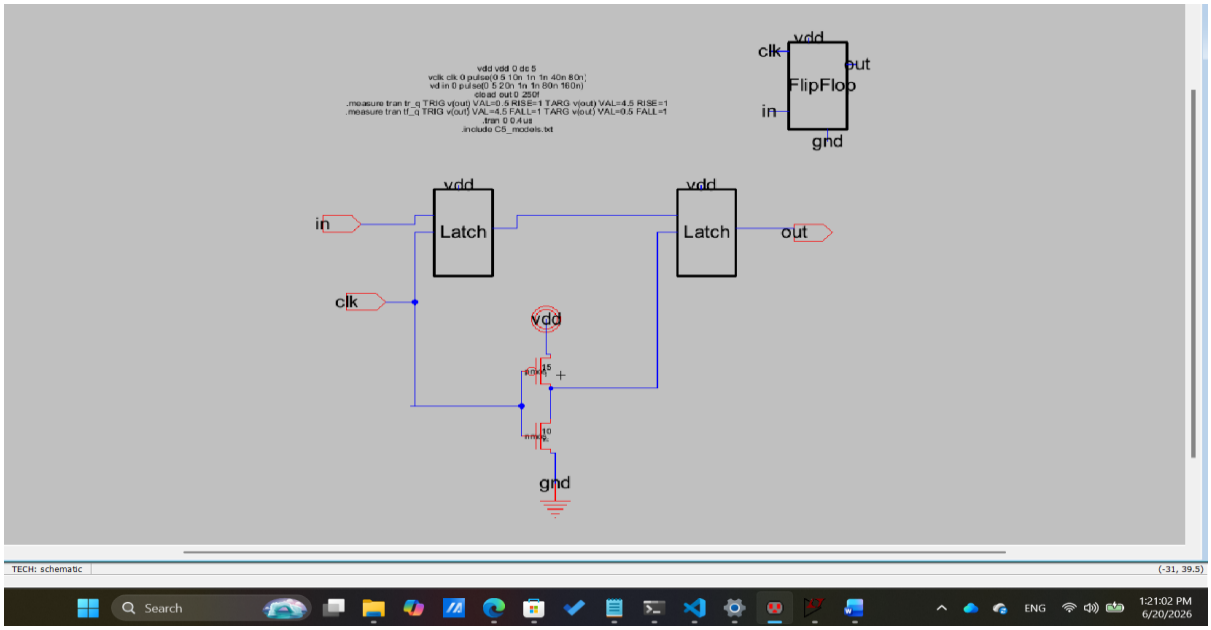



*Figure 4:Falling Edge Flip Flop Schematic*

### 2.2 Falling Edge Flip Flop Layout

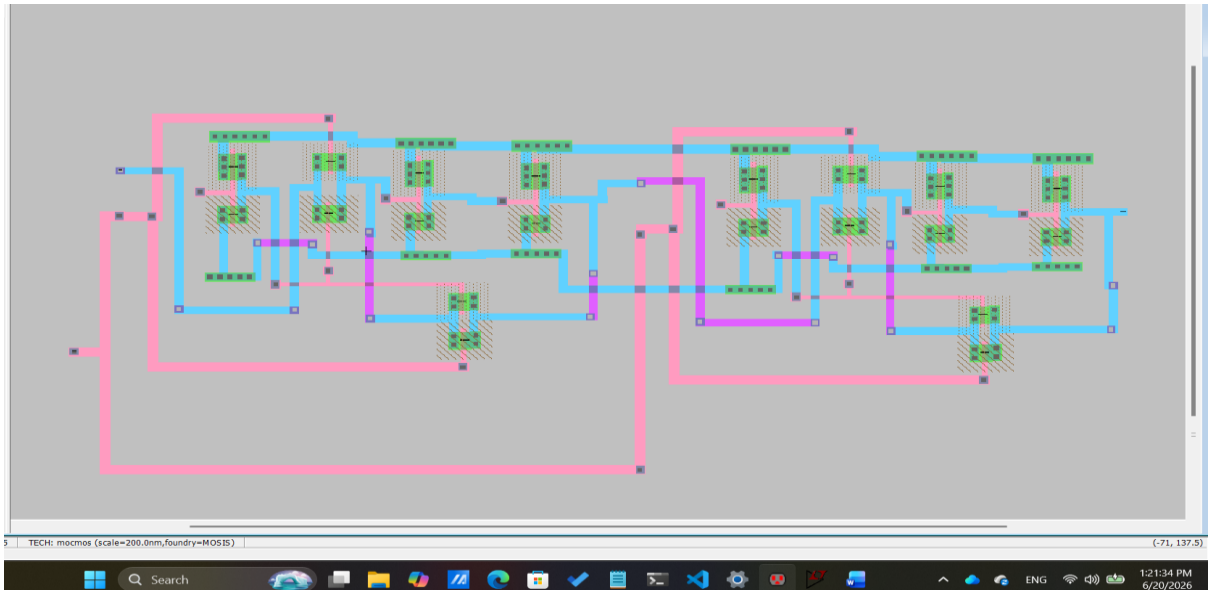


*Figure 5:Falling Edge Flip Flop Layout*

The layout of the Falling Edge D Flip-Flop was created in Electric VLSI by connecting two D Latch layouts in a master-slave configuration. Each latch was implemented using CMOS technology with PMOS transistors connected to the VDD rail and NMOS transistors connected to the GND rail. Metal-1 layers were used to route the required signals between the transistors and between the two latches.

During the layout design process, DRC checks were performed to identify spacing and overlap violations, and the layout was modified until the design rules were satisfied. The final layout correctly represents the schematic implementation of the Falling Edge D Flip-Flop and provides a valid CMOS realization of the circuit.

## 2.3 Falling Edge Flip Flop Simulation

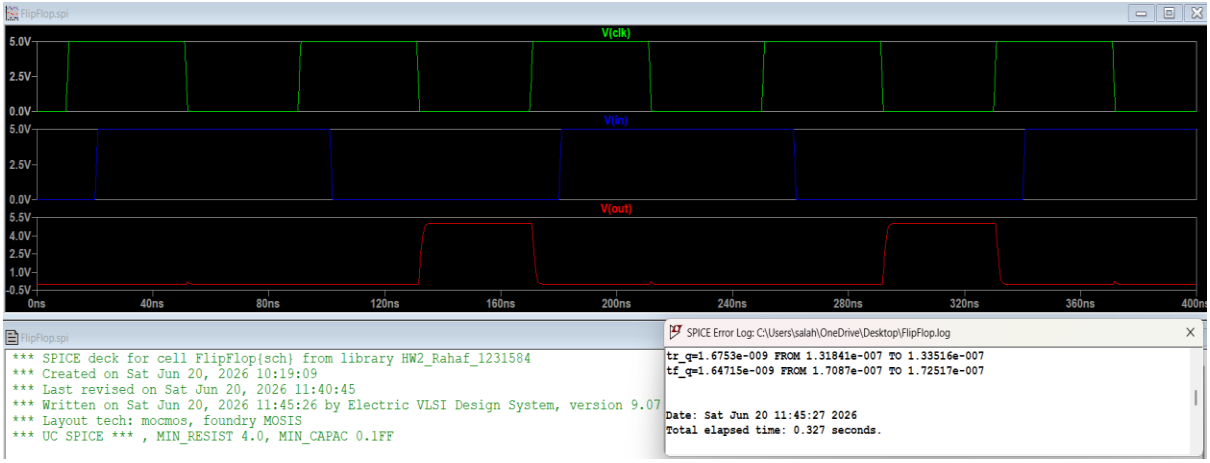



*Figure 6:Falling Edge Flip Flop Simulation*

## 2.4 Discussion

The D Flip-Flop was implemented using two D Latches connected in a master-slave configuration. The first latch operates with the clock signal (CLK), while the second latch operates with the inverted clock signal ($CL\bar{K}$). This arrangement creates a Falling Edge Triggered D Flip-Flop.

From the simulation waveform, it can be observed that the output does not change during the rising edge of the clock. Instead, the output updates only when the clock transitions from logic high to logic low (falling edge). This behavior confirms that the circuit is a Falling Edge D Flip-Flop. The output follows the value stored in the master latch and transfers it to the slave latch at the falling edge, providing stable and synchronized data storage.

The measured rise and fall times were approximately 1.68 ns and 1.64 ns, respectively, indicating fast switching performance and correct operation of the circuit.

# 3.Invertor

The CMOS inverter designed in Homework 1 was reused as a building block in this project. It was utilized in both the schematic and layout implementations to generate the complementary clock signal ($CL\bar{K}$) required for the operation of the D Latch and the Falling Edge D Flip-Flop. Reusing the inverter simplified the design process and ensured consistency between the schematic and layout implementations.

## 3.1 Invertor Schematic

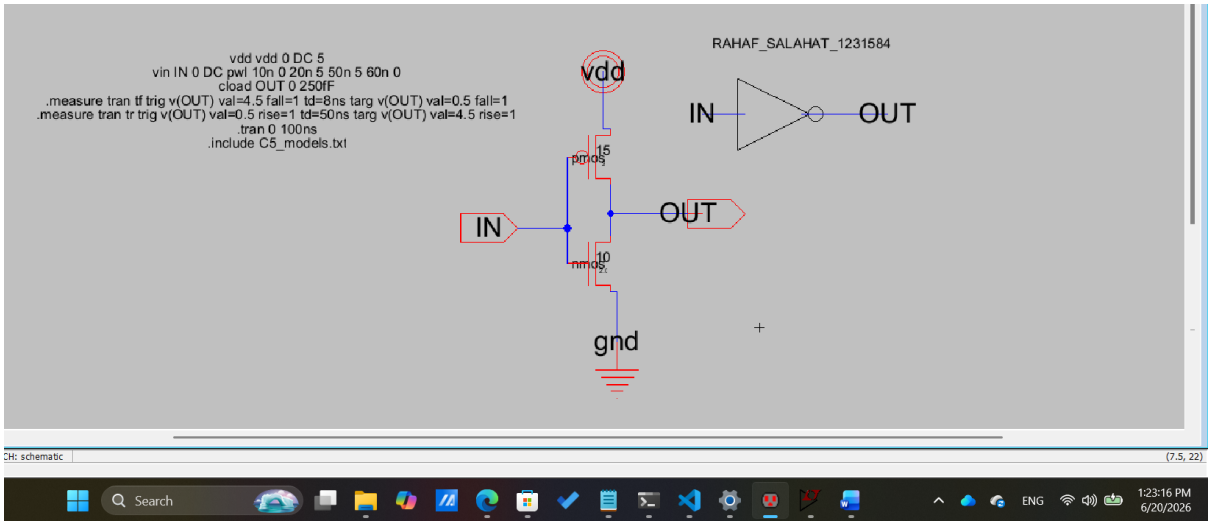



*Figure 7:3.2 Invertor Schematic*

## 3.2 Invertor Simulation

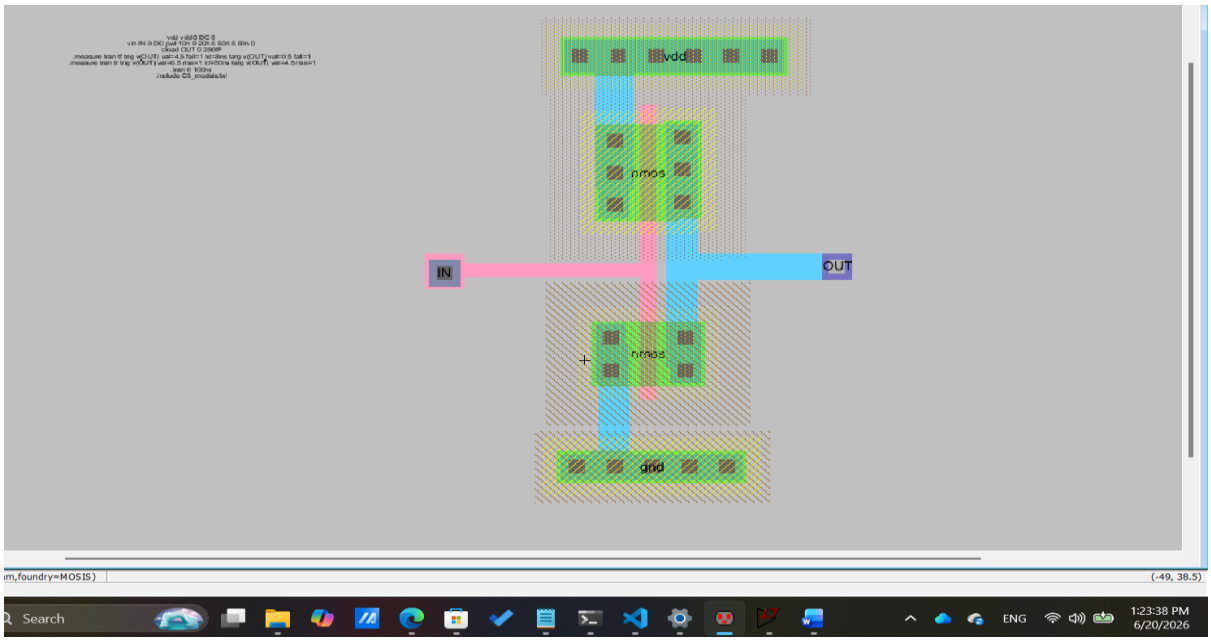



*Figure 8:Invertor Simulation*